

FAULT FINDING TOOL For Industrial Electrical Personnel

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In industrial settings, including process and power plants, electrical personnel sometimes encounter unexplained issues such as the tripping of machines, process motors, HV/LV circuit breakers, and diesel generators. These issues can occur during both the starting and running phases of electrical equipment.

Even though field control panels are provided with alarm systems or data loggers in the main control room, they may not always reveal the exact cause of a fault. In such situations, this fault finder becomes a valuable tool for field electrical personnel to identify the fault.

Presented here is a fault finder circuit based on the STM32 Blue Pill development board. Eight digital inputs and an analogue input are chosen for fault recording. The STM32 board, connected to a laptop computer with the provided Arduino program open and its serial monitor window active, will print the input number, contact status (logic 0 or 1), and its operation time in milliseconds whenever a change in contact state occurs. The 'serial print' feature of the Arduino IDE is used to record the status

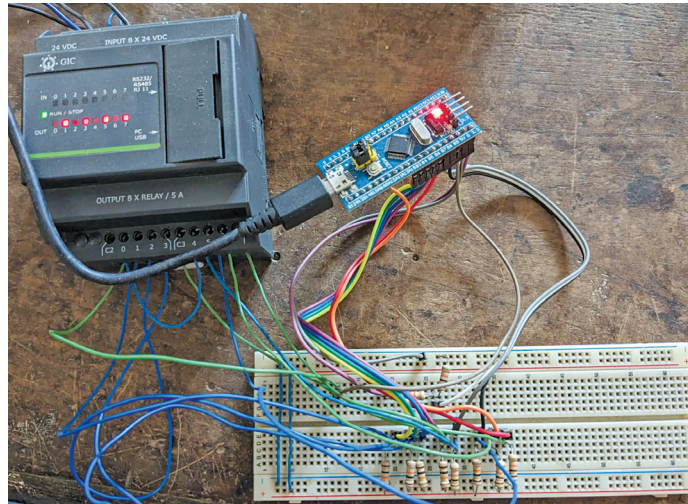


Fig. 1: Author's prototype on breadboard

and operation time of the inputs.

The purpose of this device can be explained with the help of the prototype shown in Fig. 1. If all contacts are closed/through, then the final relay-M will energise to start and run the motor/equipment continuously.

In case one or more contacts open, the relay-M will de-energise to trip the

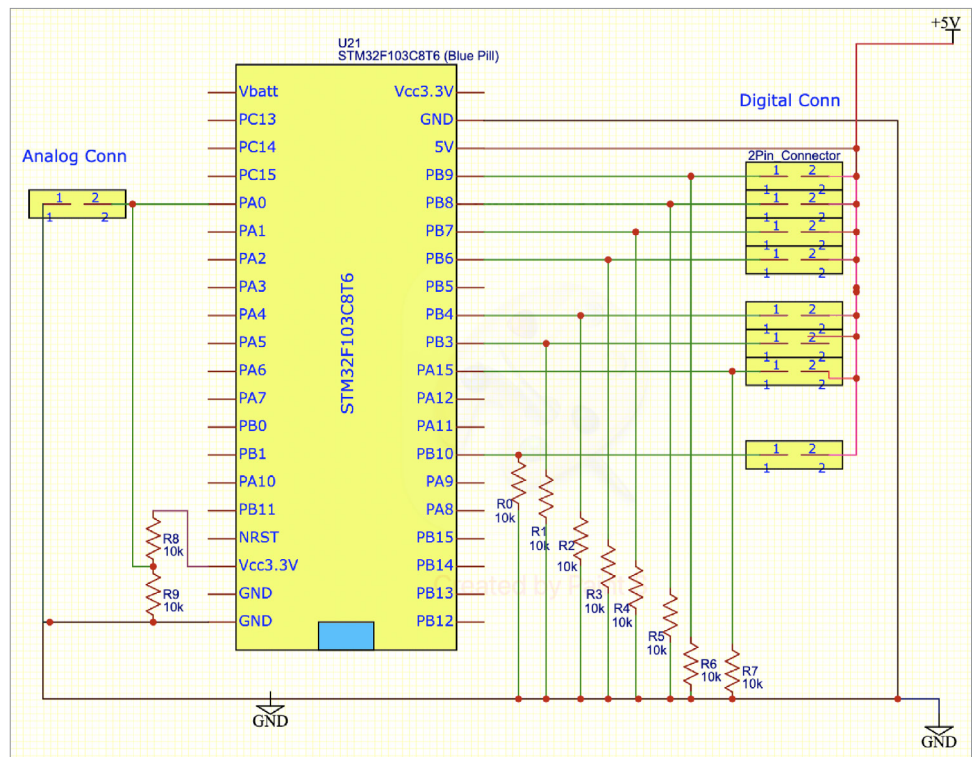


Fig. 2: Circuit diagram